MORADABAD INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SESSIONAL TEST -1st (SET-1)

Course: B.Tech. **Sem: 2nd** **Session: 2024-25** **Time: 1 hr.**
Subject: Programming For Problem Solving **Subject Code: BCS-201** **Sec: A, B, C and D** **Max. Marks: 20**

Q.No. :	1	2	3	4	5	6
CO No. :	1	1	2	2	2	3
Bloom's Level	K2	K2	K2	K3	K3	K3

CO1: To develop simple algorithms for arithmetic and logical problems.
CO2: To translate the algorithms to programs & execution (in C language).
CO3: To implement conditional branching, iteration and recursion.
Note: All sections are compulsory.

SECTION- A

1. Explain the various primitive data types on the bases of range, format specifier and memory size used in C language. **2 marks**

2. Explain the digital computer and its components with proper architecture (Block diagram). **3 marks**

3. Explain bitwise and ternary operators in detail. What is the importance of precedence order and associattivity. **3 marks**

SECTION- B

4. WAP in C to check the entered year is leap year or not. Also write the syntax of if-else statement. **4 marks**

5. WAP in C to perform arithmetic calculations (+, - , *, /) using switch statements. **4 marks**

6. WAP in C to check whether the entered positive integer number is prime or not. **4 marks**

a>b ? b>c : a : b
ll :

if(n%i==0){
c=c+1;
}

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SESSIONAL TEST -1st (SET-1) **Time: 1 hr.**

Course: B.Tech. **Sem: 1st** **Session: 2025-26** **Sec: E, F, G and H**

Subject: Programming For Problem Solving **Subject Code: BCS-101** **Max. Marks: 20**

Q.No. :	1	2	3	4	5	6
CO No. :	1	1	1	2	2	2
Bloom's Level	K2	K3	K3	K1	K3	K3

CO1 To develop simple algorithms for arithmetic and logical problems.

CO2 To translate the algorithms to programs & execution (in C language).

CO3 To implement conditional branching, iteration and recursion.

Note:

1. This paper contains two sections (A) and (B)
2. All sections are compulsory.

SECTION- A		
1.	Explain the bitwise and conditional operators in details.	**2 marks**
2.	Why break statement is used required? Write a C program to check the character is vowel or consonant.	**3 marks**
3.	Write a C program to check the entered year is leap year or not.	**3 marks**

SECTION- B		
4.	Define the following in details: **(i)** Compiler **(ii)** Interpreter **(iii)** Linker **(iv)** Loader	**4 marks**
5.	Draw the architecture of Digital Computer System and its components in details.	**4 marks**
6.	Write a Program to check whether the entered number is prime number or not.	**4 marks**

MORADABAD INSTITUTE OF TECHNOLOGY
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SESSIONAL TEST -2nd (SET-2)

Course: B.Tech. **Session: 2025-26** **Time: 1 hr.**
Sem: 1st **Subject Code: BCS-101** **Sec: E, F, G and H**
Subject: Programming For Problem Solving **Max. Marks: 20**

Q.No. :	1	2	3	4	5	6
CO No. :	4	4	4	5	5	3
Bloom's Level	K3	K2,K3	K1,K3	K2	K2	K3

CO3: To implement conditional branching, iteration and recursion.
CO4: To decompose a problem into functions and synthesize a complete program using divide and conquer approach.
CO5: To use arrays, pointers and structures to develop algorithms and programs.

Note:
1. This paper contains two sections (A) and (B)
2. All sections are compulsory.

	SECTION- A	
1.	Write a program to search an element in an array using linear search method.	**2 marks**
2.	Write a program to sort elements of the 1D array using selection sort method and explain it with an example.	**3 marks**
3.	Define function and write a program to swap two numbers using call by value and call by reference method.	**3 marks**

	SECTION- B	
4.	Discuss dynamic memory allocation in details. Also explain malloc(), calloc(), realloc() and free() functions with syntax.	**4 marks**
5.	Explain Macro and its types with suitable example of each in details.	**4 marks**
6.	Write a program to multiply two matrices of **nxn** order.	**4 marks**

MORADABAD INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SESSIONAL TEST -2nd (SET-2)

Course: B.Tech. Sem: 2nd Session: 2024-25
Subject: Programming For Problem Solving Subject Code: BCS-201

Q.No. :	1	2	3	4	5
CO No. :	3	3	3	3	3
Bloom's Level	K3	K2	K3	K3	K2

CO3: To implement conditional branching, iteration and recursion.
CO4: To decompose a problem into functions and synthesize a complete program using divide and conquer approach.
CO5: To use arrays, pointers and structures to develop algorithms and programs.

Note:
1. This paper contains two sections (A) and (B)
2. All sections are compulsory.

SECTION- A

1. WAP in C to add two nxn matrices and print the sum matrix.
2. Explain the difference between structure and union in detail.
3. WAP in C to find the factorial of positive integer number using recursive function.

SECTION- B

4. WAP in C to sort the elements of the array using bubble sort method.
5. Explain in details the Macro and its types with suitable example.
6. Explain malloc(), calloc(), realloc() and free() functions used in Dynamic Memory Allocation with suitable example. 4 marks

nr (fib(num));
factorial
nr (fib n-2);